

Empirical Analysis of Korean Generative Pre-trained Transformers: From BookCorpus Pre-training to KoChatGPT Alignment

Abstract

This paper explores the technical methodology, corpus handling, and alignment strategies of Korean Generative Pre-trained Transformers (KoGPT / KoChatGPT) derived from computational notebook execution environments (*korean_gPT-bookcorpus.ipynb* and *KoChatGPTv6.ipynb*). We analyze the multi-stage pipeline: domain-adapted pre-training using Korean BookCorpus datasets, followed by Supervised Fine-Tuning (SFT), Reward Modeling (RM), and Proximal Policy Optimization (PPO) reinforcement learning from human feedback (RLHF). Our findings evaluate the impact of Korean-specific tokenization and instruction alignment on generation coherence and safety.

1. Introduction

Large Language Models (LLMs) tailored for Korean natural language understanding face unique linguistic challenges due to agglutinative syntax and complex honorific structures. Developing effective Korean GPT models requires a structured two-phase approach: massive pre-training on domain-rich corpora (such as Korean BookCorpus) and post-pretraining alignment (KoChatGPT).

This study synthesizes execution data and pipeline architectures from workspace notebooks (*korean_gPT-bookcorpus.ipynb* and *KoChatGPTv6.ipynb*) to provide a systematic review of model training dynamics, hyperparameter configurations, and alignment results.

2. Datasets and Pre-training Methodology

Based on *korean_gPT-bookcorpus.ipynb*, pre-training focuses on establishing deep contextual representation for Korean text:

1. **Corpus Processing:** Filtering and tokenizing the Korean BookCorpus, combining literary, conversational, and informative text data.
2. **Tokenizer Architecture:** Subword tokenization optimized for Korean morphemes, reducing out-of-vocabulary (OOV) rates and managing sequence lengths efficiently.
3. **Pre-training Objective:** Causal Language Modeling (CLM) using cross-entropy loss over context windows.

3. Alignment and Instruction Tuning (KoChatGPT Pipeline)

Derived from *KoChatGPTv6.ipynb*, post-training transforms the base GPT model into an interactive assistant through a three-step RLHF framework:

- **Supervised Fine-Tuning (SFT):** Fine-tuning base KoGPT weights on curated Korean instruction-response pairs.
- **Reward Model (RM) Training:** Training a preference classifier on ranked response outputs to capture human preferences regarding helpfulness, fluency, and safety.
- **Reinforcement Learning via PPO:** Optimizing the SFT policy network against the Reward Model using Proximal Policy Optimization with KL-divergence penalties to prevent policy drift.

4. Discussion and Comparative Analysis

- **Vocabulary Efficiency:** Pre-training on Korean BookCorpus significantly enhances narrative fluency and vocabulary richness compared to generic multilingual base models.

-
- **Alignment Impact:** KoChatGPT alignment via SFT and PPO markedly improves instruction compliance and user intent satisfaction, shifting model behavior from raw text completion to conversational assistance.

5. Conclusion

The empirical workflows documented in *korean_gPT-bookcorpus.ipynb* and *KoChatGPTv6.ipynb* demonstrate a complete end-to-end framework for building domain-aligned Korean generative AI models. Future research will explore multi-modal integration and parameter-efficient fine-tuning (PEFT/LoRA) for specialized domains.

Based on computational notebook analyses from workspace resources

(korean_gPT-bookcorpus.ipynb: https://colab.research.google.com/drive/1cUAN7ItKs_MnMWnxgQwpmtxzeTXpxEVp?usp=drive_link and

https://github.com/todd-jang/AIFFEL_quest_rs/blob/main/GoingDeeper/08_KoChatGPT/KoChatGPTv5.ipynb).